

Center for Operational Oceanographic Products and Services
NATIONAL OCEAN SERVICE

CONRAD BLUCHER INSTITUTE

An efficient AI model for automatic QA/QC flag detection for water level data

Presenters: Lindsay Abrams[2] & Thanh-Dung Le[1]

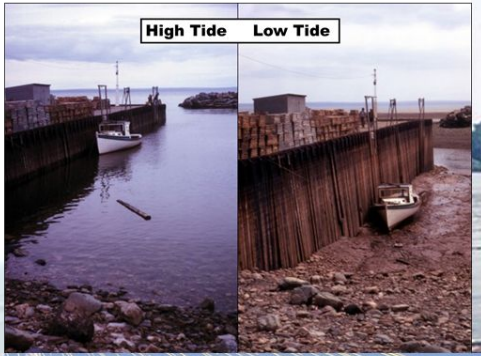
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Ocean Modeling GitHub Community
October 1st, 2026

About Our Data

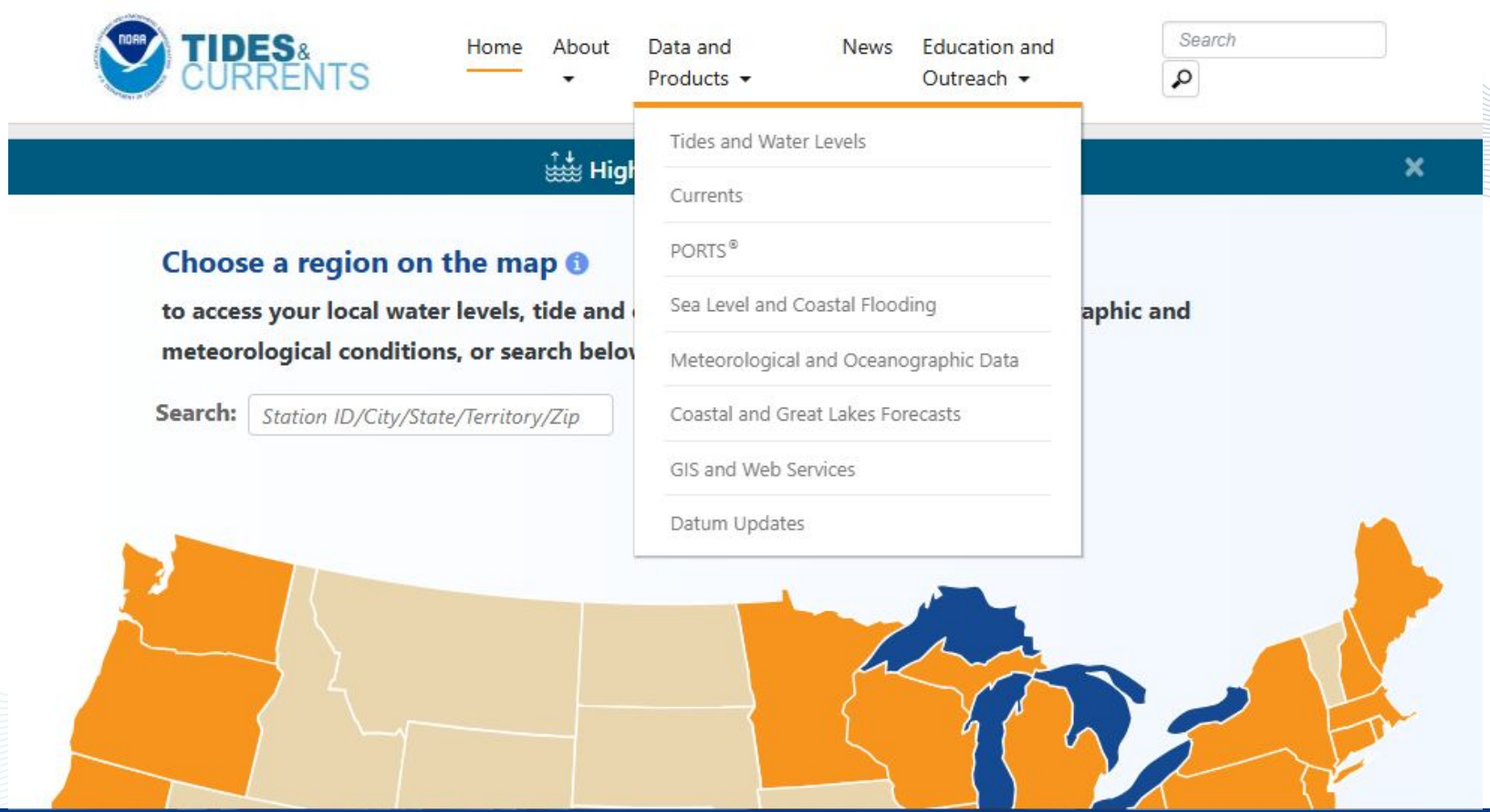
- Authoritative source for tides, water levels, currents, and other coastal ocean and meteorological information.
- 300+ water level stations on the U.S. coasts and Great Lakes, including met sensors and current meters, concentrated in busy seaports.
- Provide historic and real-time data, forecasts, predictions, and scientific products that protect life, the economy, and the environment on the coast.





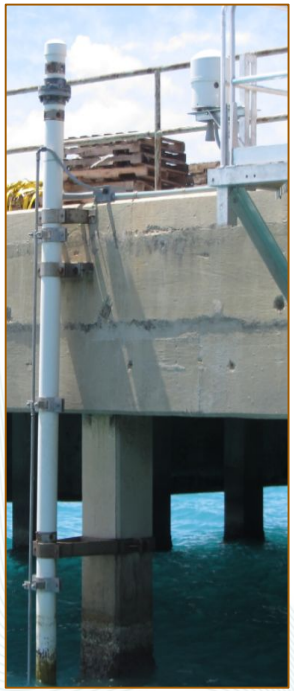


About Our Products



Existing QC process

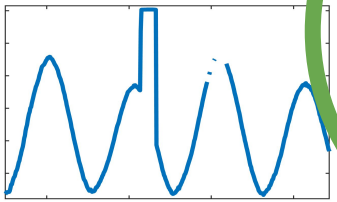
Primary + Backup
water level sensors



Onsite QC

Outlier detection
6-minute averaging

Preliminary Data

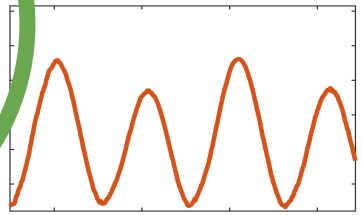


Manual processing
and verification



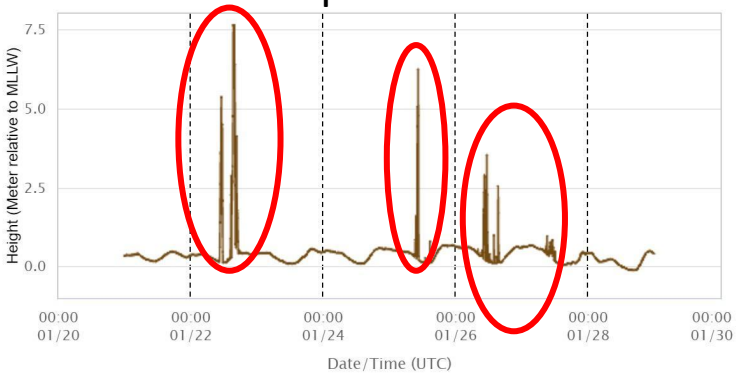
-Review QC flags
-Manually remove bad data
-Fill remaining gaps with backup, neighbor & tide prediction data

Verified data

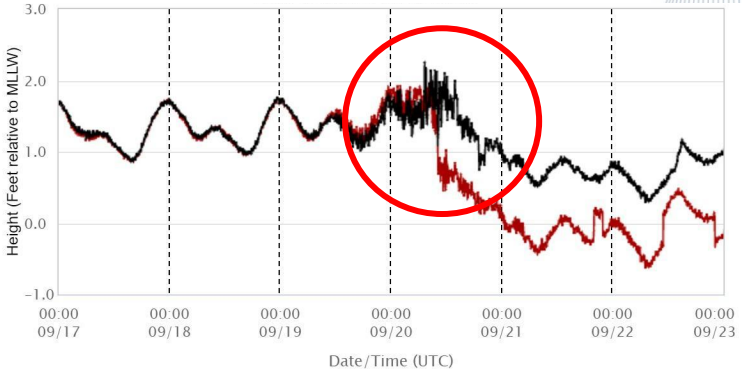


Data Quality Issues

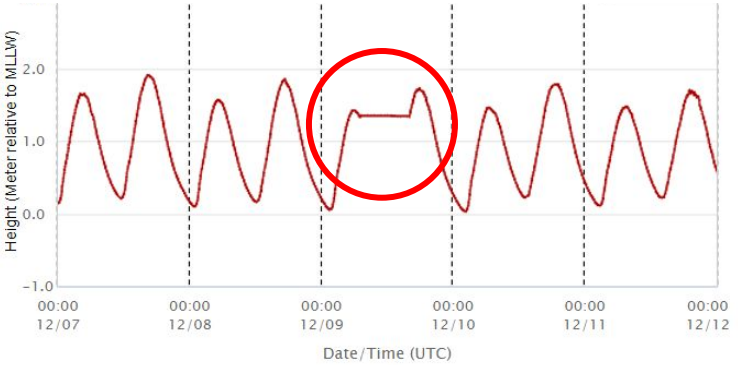
Spikes



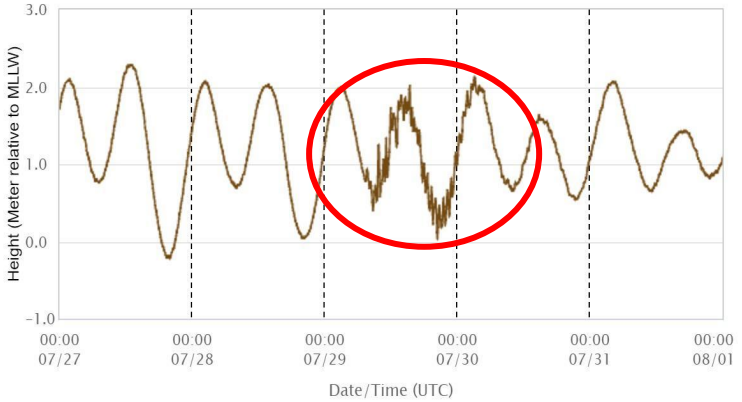
Data Shifts



Flats



Rare events (good data)



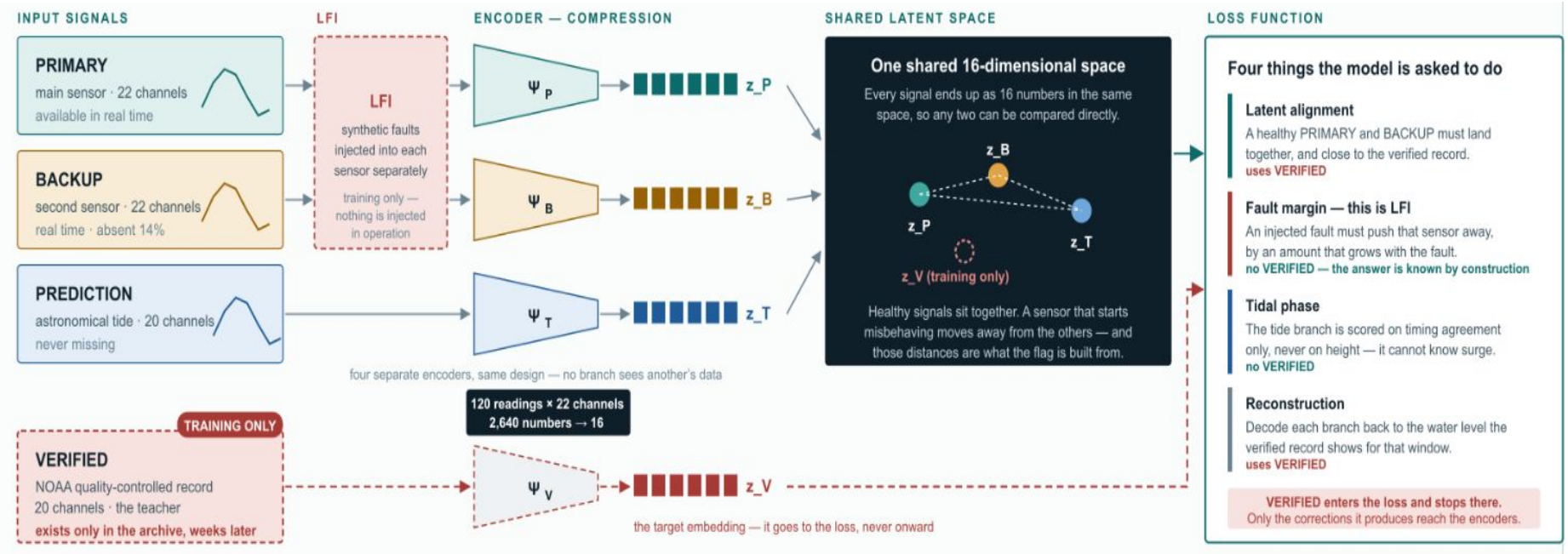
Research Objectives

Develop an AI-enabled quality control system based on decades of QC'd water level data that improves accuracy and efficiency in generating quality-controlled water level observations. The system should be able to:

1. Classify 6-minute water level preliminary sensor data as good or bad at a high accuracy and
2. Replace and/or fill gaps in the resultant time series.

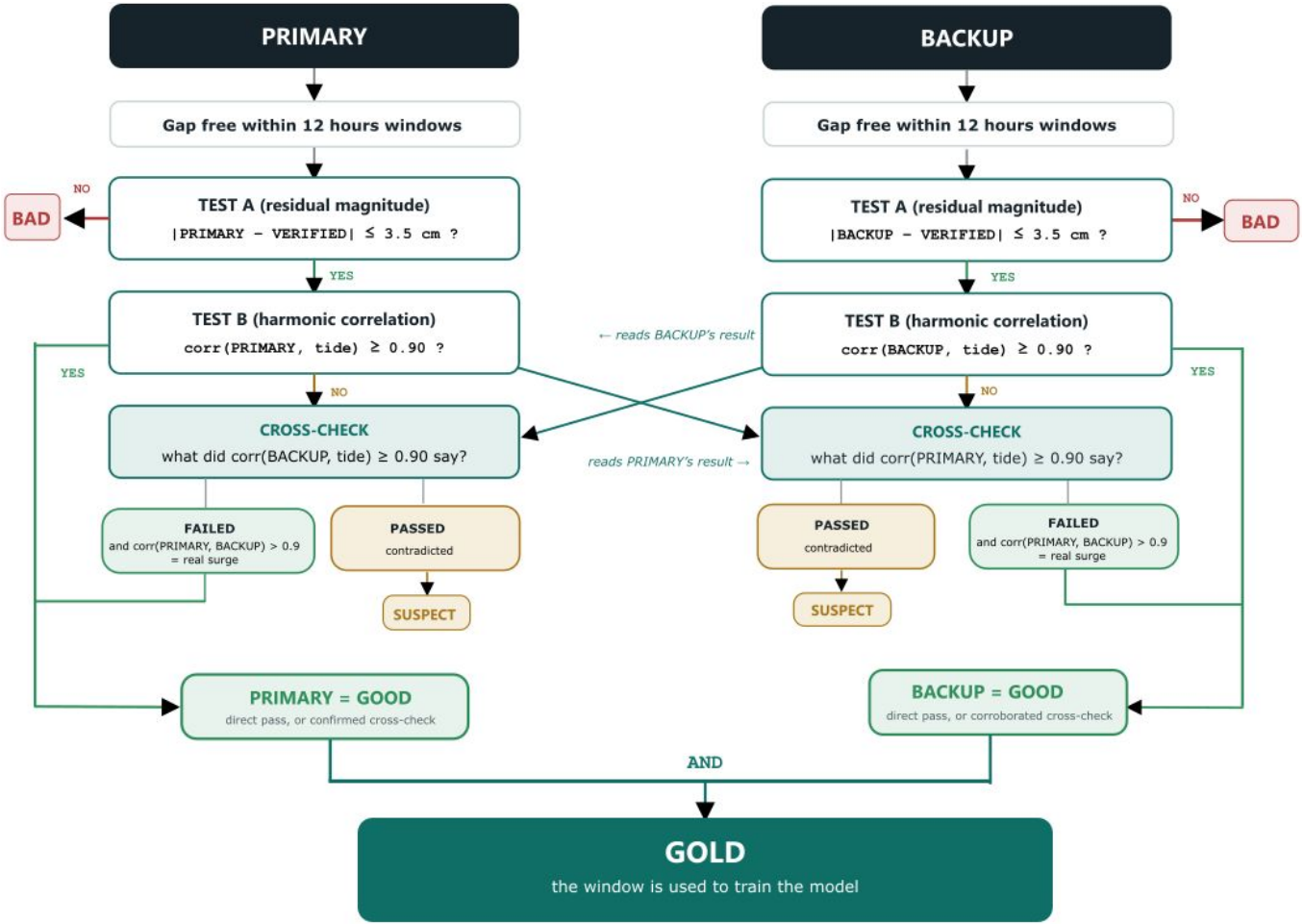
Method - JEPA

The main objective is train JEPA (**J**oint-**E**mbedding **P**redictive **A**rchitecture) learn to have a latent space that can capture and learn good physical signal from PRIMARY, BACKUP from the NOAA's VERIFIED.



The input features from each sensor including different characteristics, as following:
Group 1: The actual value (PRIMARY, BACKUP, TIDAL PREDICTION)
Group 2: The local dynamics from last 30 minutes (the _DIFF1, ROLL_STD_*, ROLL_RANGE_* features)
Group 3: The shared context (cyclic time, tidal phase, station metadata, geography)

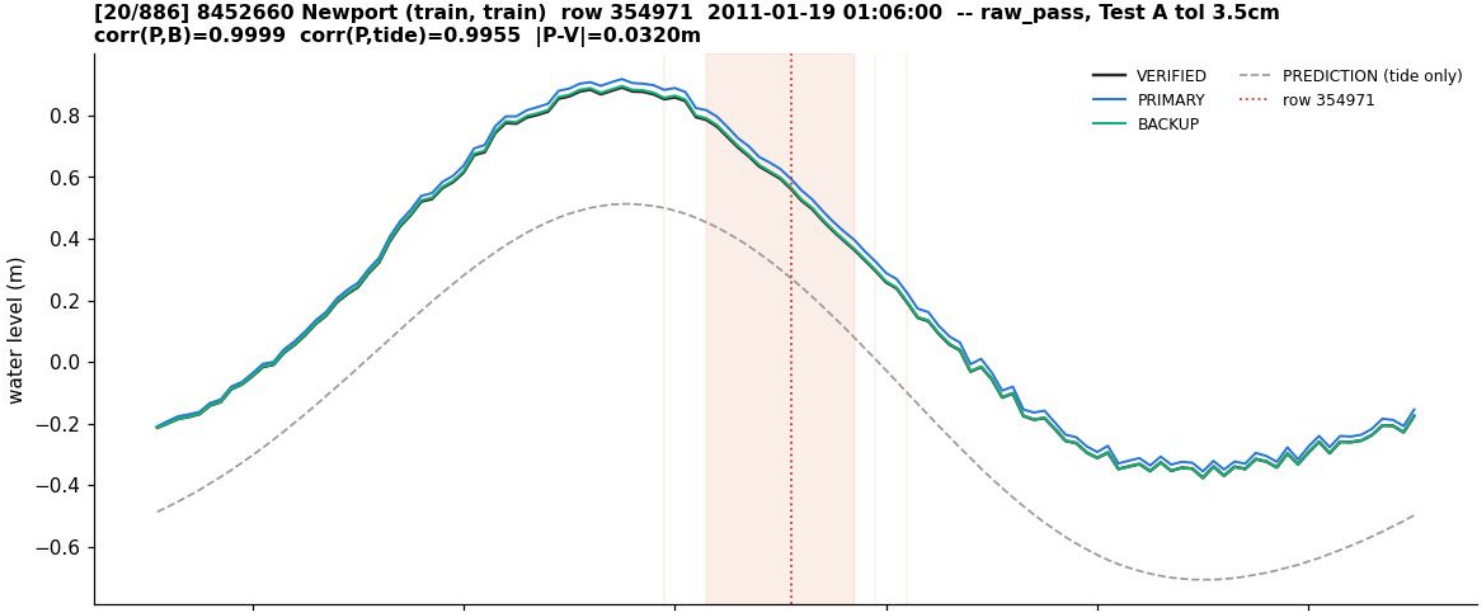
Method - Preparing GOLD Data for Training



Method - Training GOLD Data

29 stations, 7 Regions Covered: Pacific Island, Northeast, Mid-Atlantic, Southeast, Gulf, West, and Caribbean.

	Train (2007-2016)	Validation (2017-2018)	Test (2019-2020)
Data (12h gap-free)	28,879,490	6,376,000	3,574,600
GOLD Data	20,860,494 (61.49%)	4,545,296 (65.33%)	2,674,828 (62.83%)



Experiment Result - Reconstruction Performance

11 unseen stations from 4 different geography coasts.

Northeast: Montauk

Southeast: Trident Pier

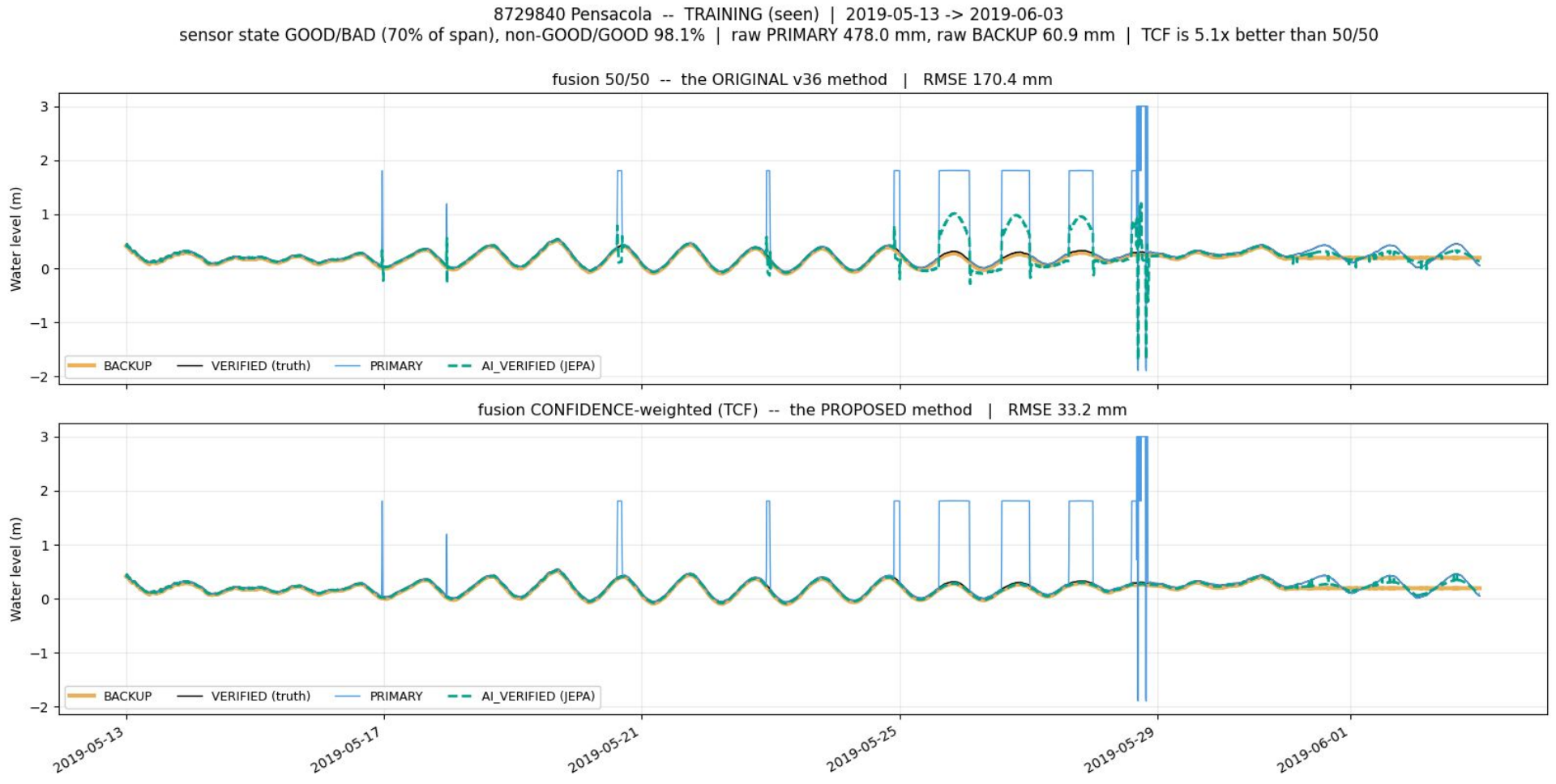
Gulf: St Petersburg, Panama City, Dauphin Island, Lake Charles, Bulk Terminal, Galveston Bay Entrance

West: Charleston, Sand Point, and King Cove

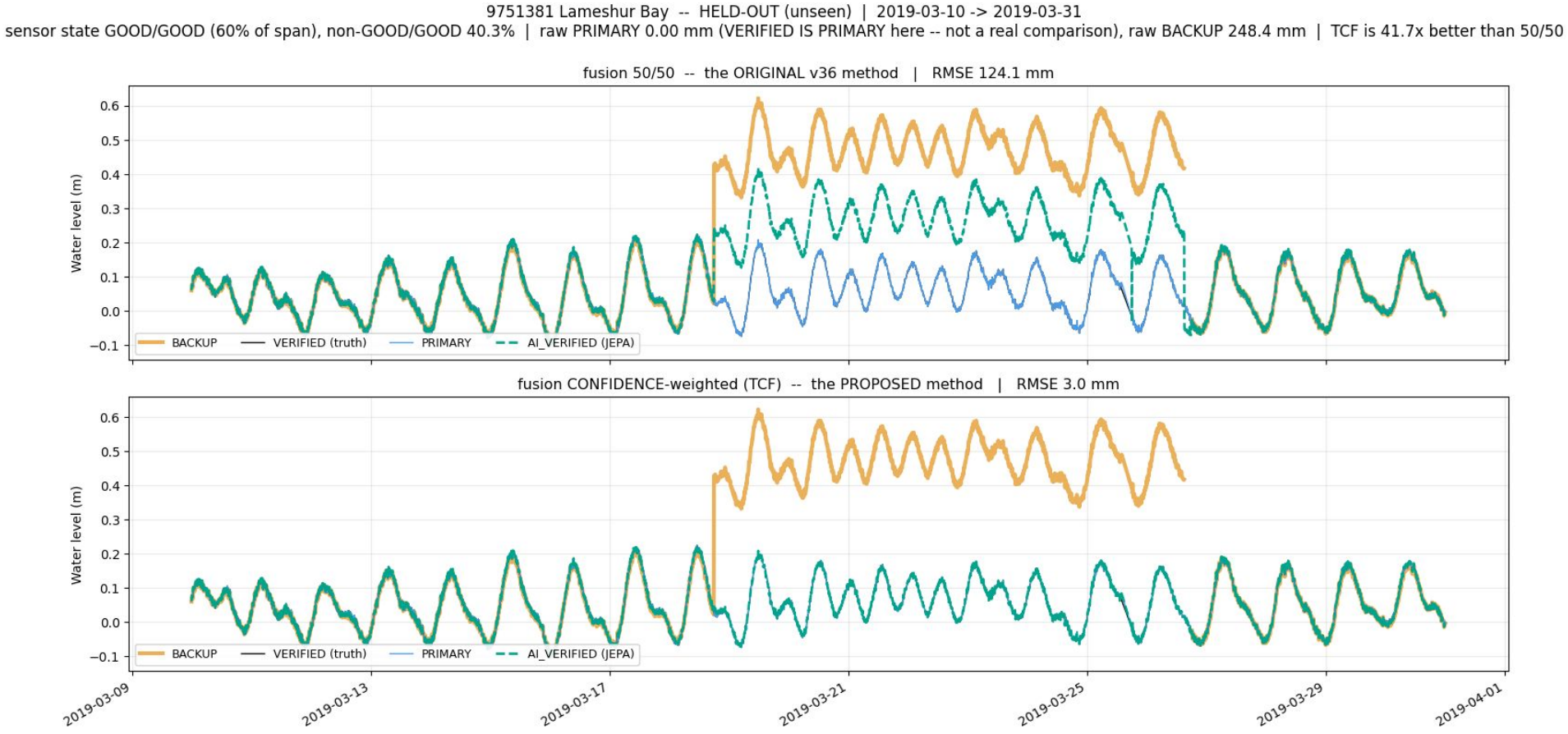
held-out unseen, 11 unseen stations	RMSE (m)
All rows	0.035
P==V (PRIMARY followed VERIFIED)	**0.017**
Best validation loss	0.034

Total the 1,112,400 rows from test split (2019-2020), 11 unseen stations, NOAA flags only 22,259 as BAD (2.0%). The other 98% are GOOD.

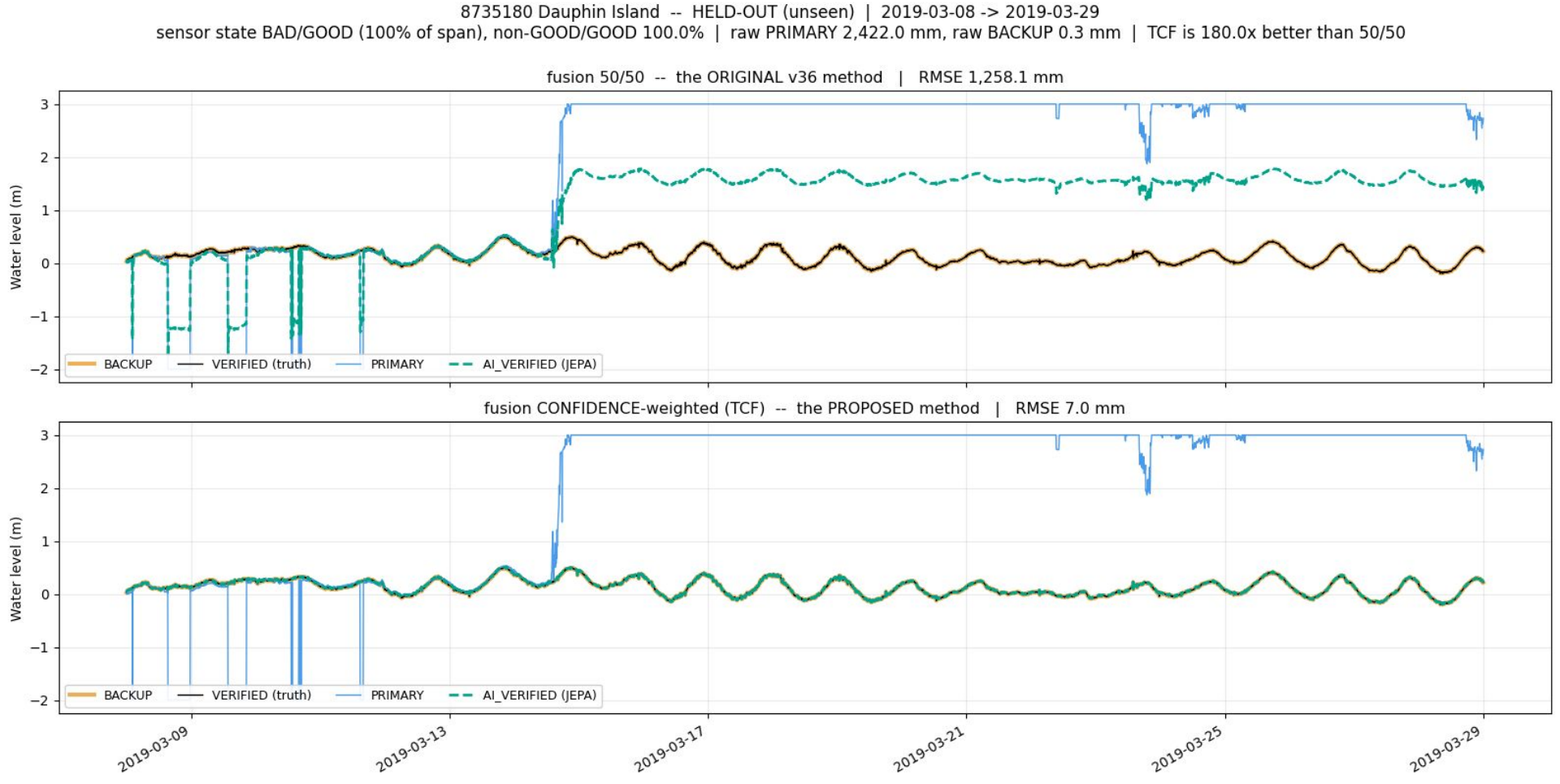
Experiment Result - Reconstruction Performance



Experiment Result - Reconstruction Performance



Experiment Result - Reconstruction Performance



Surprisingly !!!!!

One JEPA model jointly supports both QC flagging and AI-VERIFIED reconstruction.

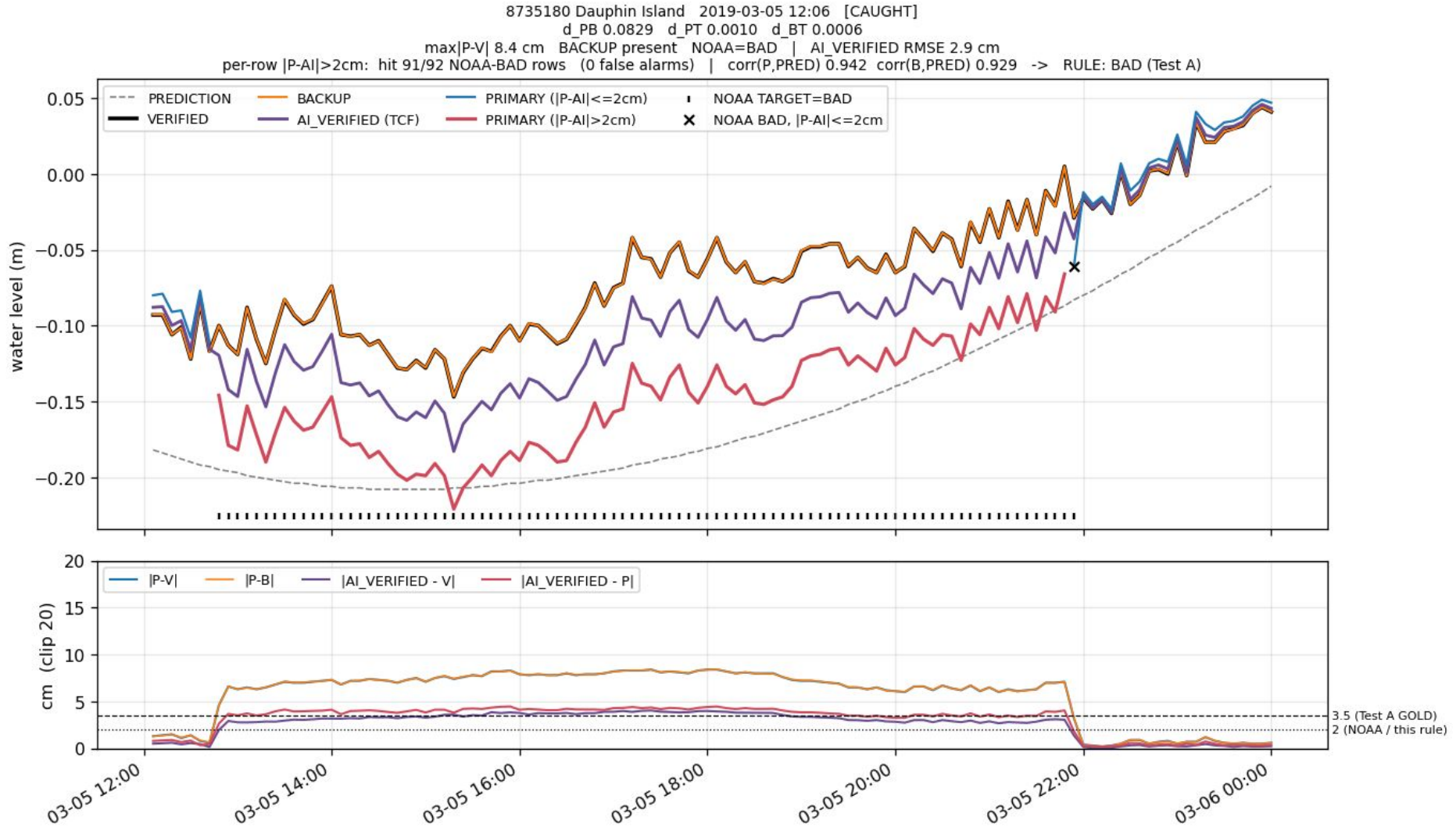
A single JEPA latent-space model supports both operational goals:

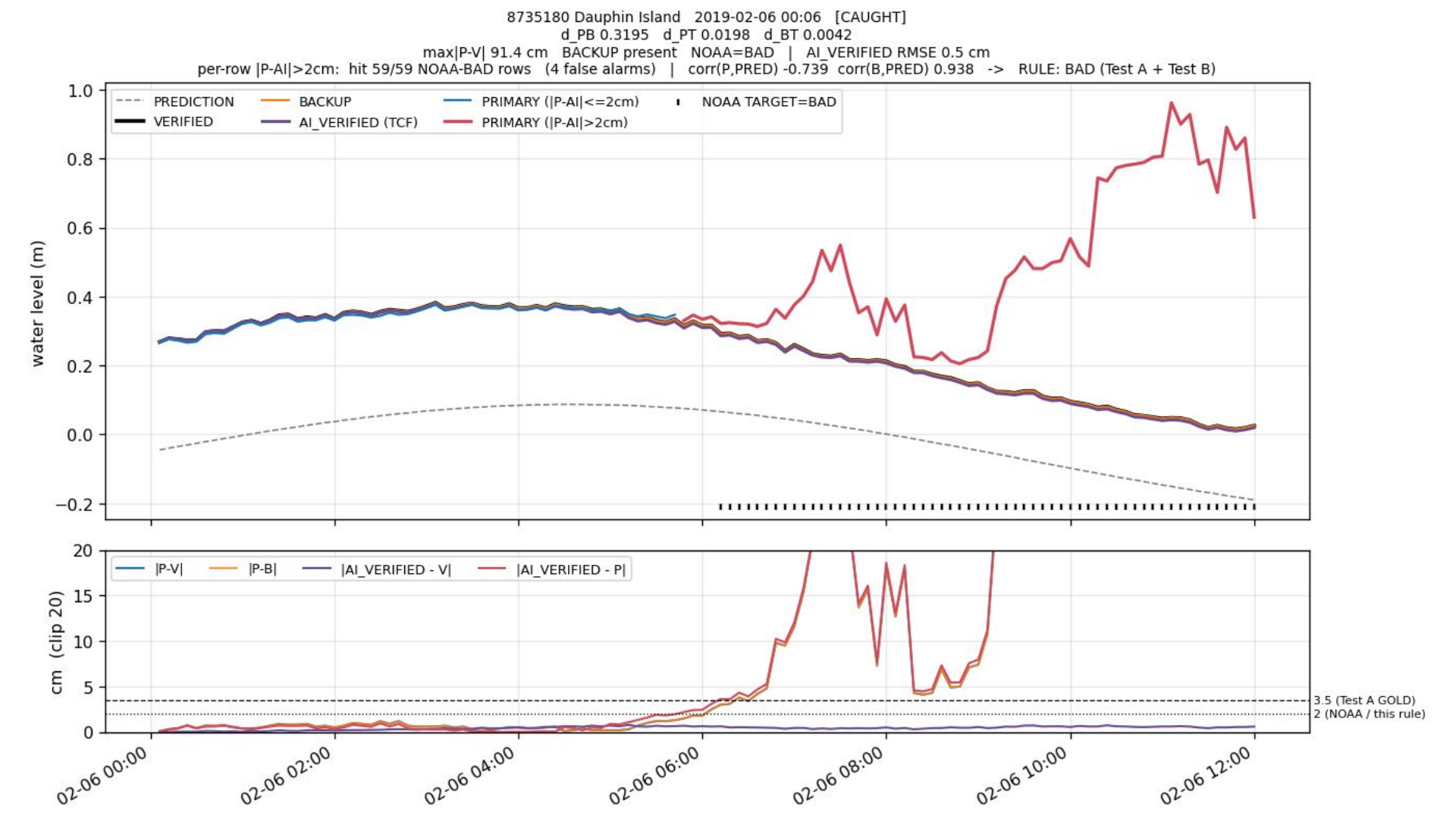
- QC: identify unphysical/bad water-level observations
- Reconstruction: generate a cleaned AI-VERIFIED water level

Experiment Result - Detection Performance

11 unseen stations, NOAA flags only BAD (2.0%), GOOD (98%).

- Window/episode-level recall: 262/273 = 95.6% of fault windows caught
- 11 windows missed (4.4%): Panama City spikes with BACKUP absent, small sustained bias at Dauphin/Galveston).


8735180 Dauphin Island 2019-03-05 12:06 [CAUGHT]
d_PB 0.0829 d_PT 0.0010 d_BT 0.0006
max|P-V| 8.4 cm BACKUP present NOAA=BAD | AI_VERIFIED RMSE 2.9 cm
per-row |P-AI|>2cm: hit 91/92 NOAA-BAD rows (0 false alarms) | corr(P,PRED) 0.942 corr(B,PRED) 0.929 -> RULE: BAD (Test A)
PREDICTION
VERIFIED
BACKUP
AI_VERIFIED (TCF)
PRIMARY (|P-AI|<=2cm)
PRIMARY (|P-AI|>2cm)
NOAA TARGET=BAD
NOAA BAD, |P-AI|<=2cm
water level (m)
0.05
0.00
−0.05
−0.10
−0.15
−0.20
|P-V|
|P-B|
|AI_VERIFIED - V|
|AI_VERIFIED - P|
cm (clip 20)
20
15
10
5
0
3.5 (Test A GOLD)
2 (NOAA / this rule)
03-05 12:00
03-05 14:00
03-05 16:00
03-05 18:00
03-05 20:00
03-05 22:00
03-06 00:00


8735180 Dauphin Island 2019-02-06 00:06 [CAUGHT]
d_PB 0.3195 d_PT 0.0198 d_BT 0.0042
max|P-V| 91.4 cm BACKUP present NOAA=BAD | AI_VERIFIED RMSE 0.5 cm
per-row |P-AI|>2cm: hit 59/59 NOAA-BAD rows (4 false alarms) | corr(P,PRED) -0.739 corr(B,PRED) 0.938 -> RULE: BAD (Test A + Test B)
PREDICTION
VERIFIED
BACKUP
AI_VERIFIED (TCF)
PRIMARY (|P-AI|<=2cm)
PRIMARY (|P-AI|>2cm)
NOAA TARGET=BAD
water level (m)
|P-V|
|P-B|
|AI_VERIFIED - V|
|AI_VERIFIED - P|
cm (clip 20)
3.5 (Test A GOLD)
2 (NOAA / this rule)
02-06 00:00
02-06 02:00
02-06 04:00
02-06 06:00
02-06 08:00
02-06 10:00
02-06 12:00

Computation Cost

Cost	Value
Checkpoint load (once)	150 s (bootstrap training notebook; 0.3 MB file, 74,225 parameters)
Per-request: 1-month	8.8 ± 0.4 seconds total
— Data preparation	8.5 ± 0.4 s (97% of the cost — CSV read + calibration + windowing)
— Model inference	0.2 ± 0.03 s (2% — the neural network itself)
— Scoring	0.1 s
Temporary storage	1.2 MB per request
GPU required?	No — CPU-only, any modern x86-64
RAM	< 4 GB

Scaling:

- 57 stations × 1 month = ~8 minutes sequential
- 57 stations × 12 months = ~1.7 hours sequential
- Fully parallelizable (each station independent)

The bottleneck is data preparation (reading ~200 MB of CSVs per station), not the model. The model itself is tiny. The trained checkpoint jepa_36v3_best.pt is 317 KB.

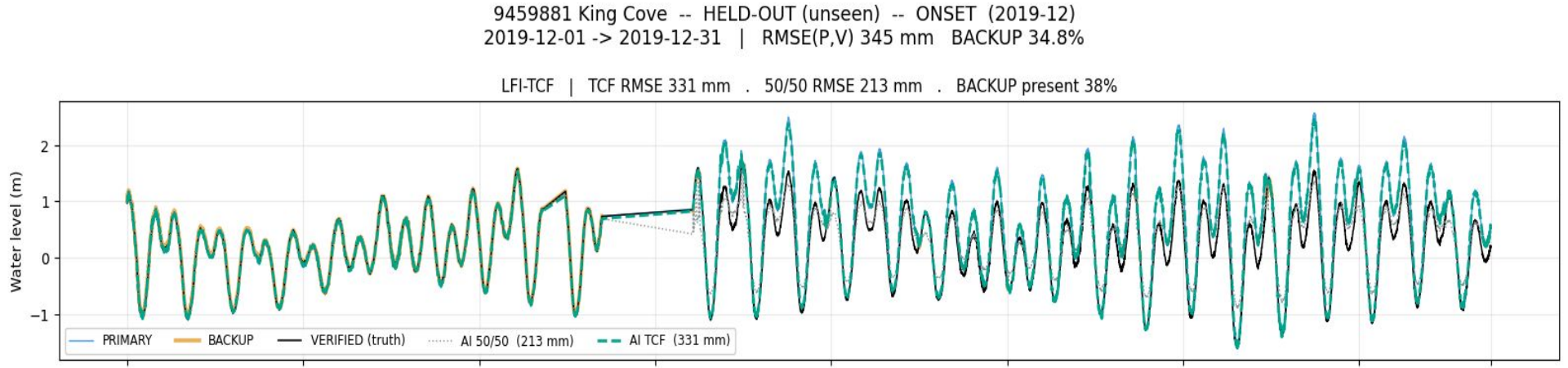
Conclusion

JEPA latent space-based model perform well to identify unphysical data:

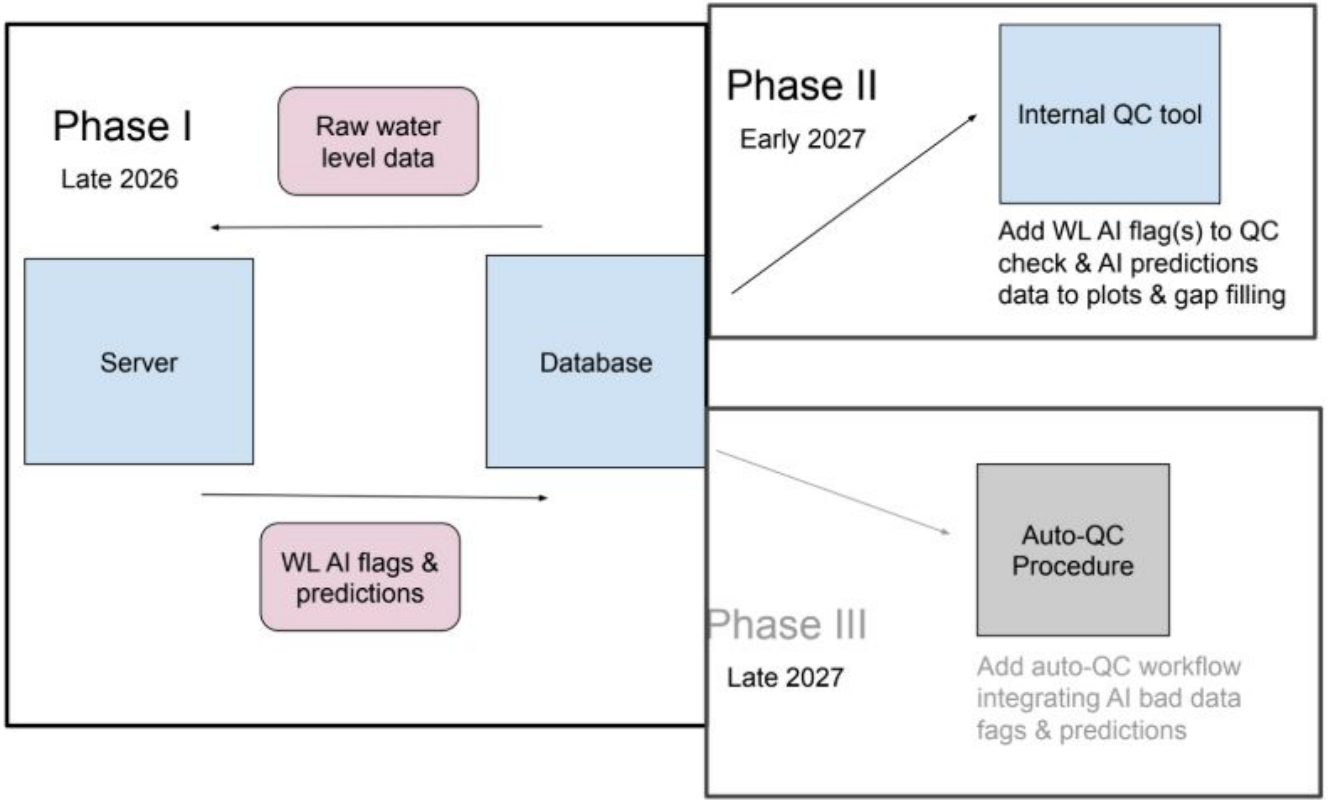
- Effectively reconstruct to cleaned AI_VERIFIED (noisy primary + available BACKUP)
- Correctly flag 262/273 = 95.6% of fault windows (12 hours gap free)

Next step:

- Improve prediction in case fault PRIMARY, missing BACKUP - **Challenging**
- Flag GOOD/BAD 6 minutes water level point (not the whole 12 hours windows)



Implementation in Operations



Thank you!







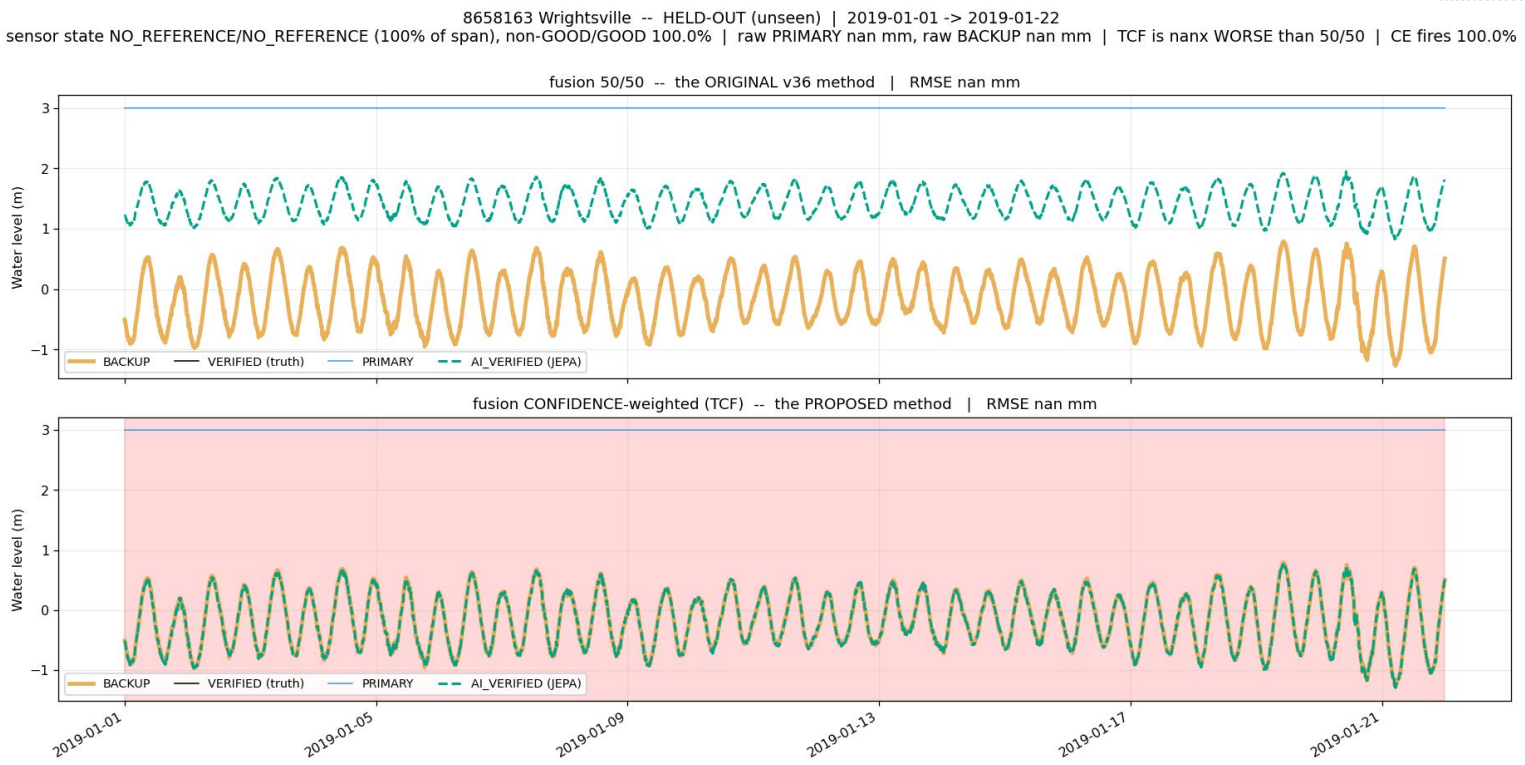
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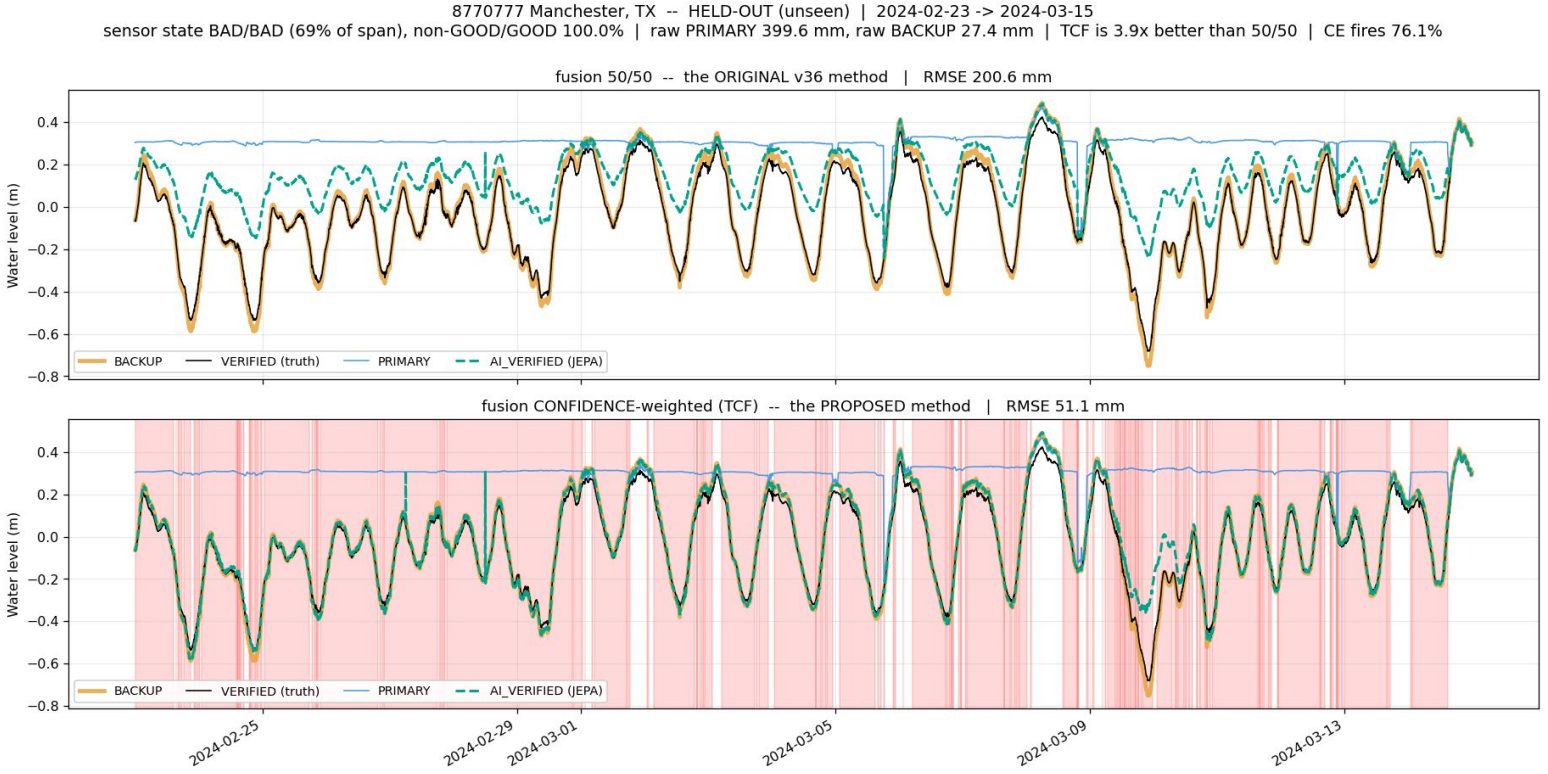
Wrightsville Beach, NC

Flat-lined primary data



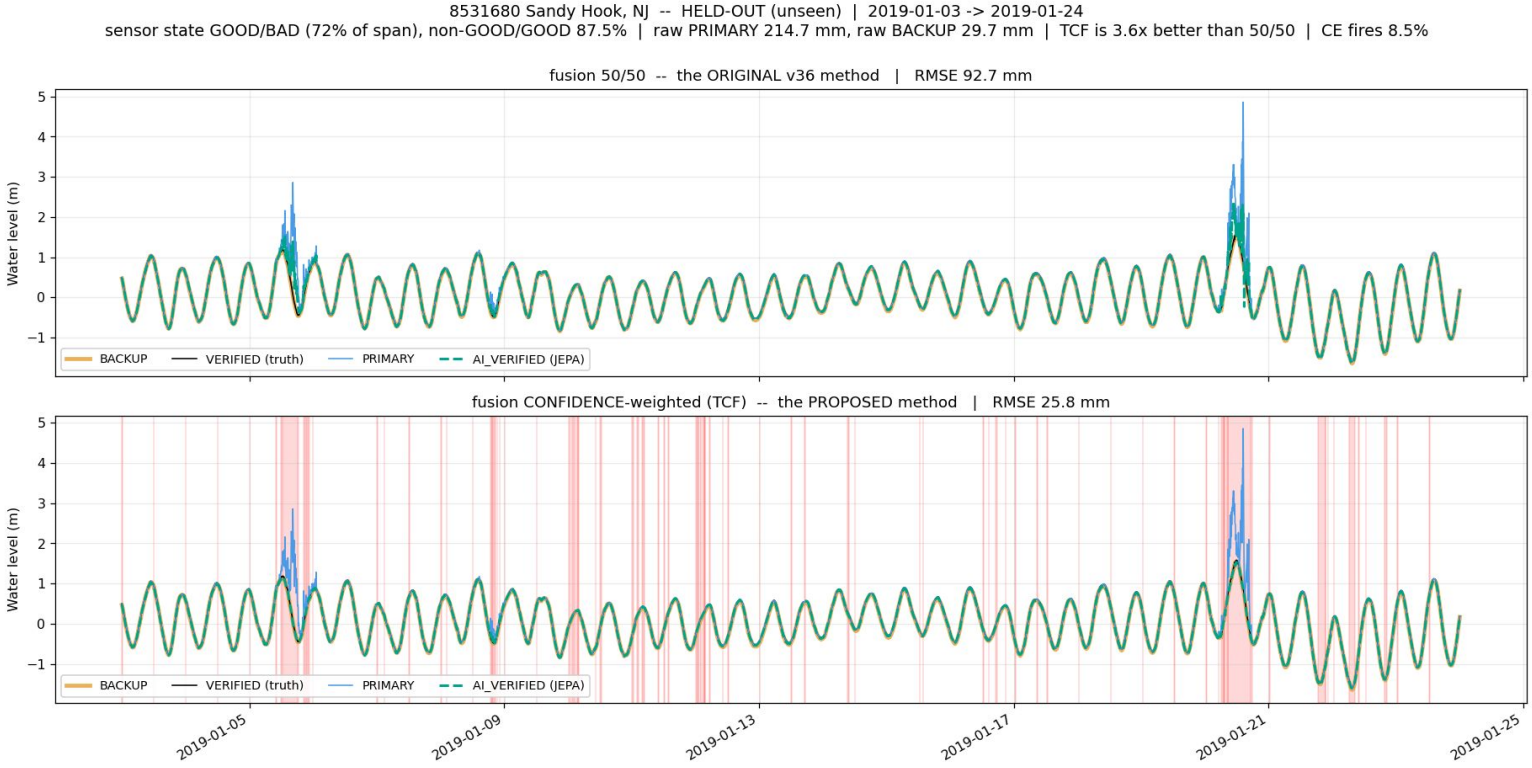
Manchester, TX

Flat-lined except for highest tides



Sandy Hook, NJ

Irregular spikes/splashy



Dauphin Island, AL

Spikes & flats

